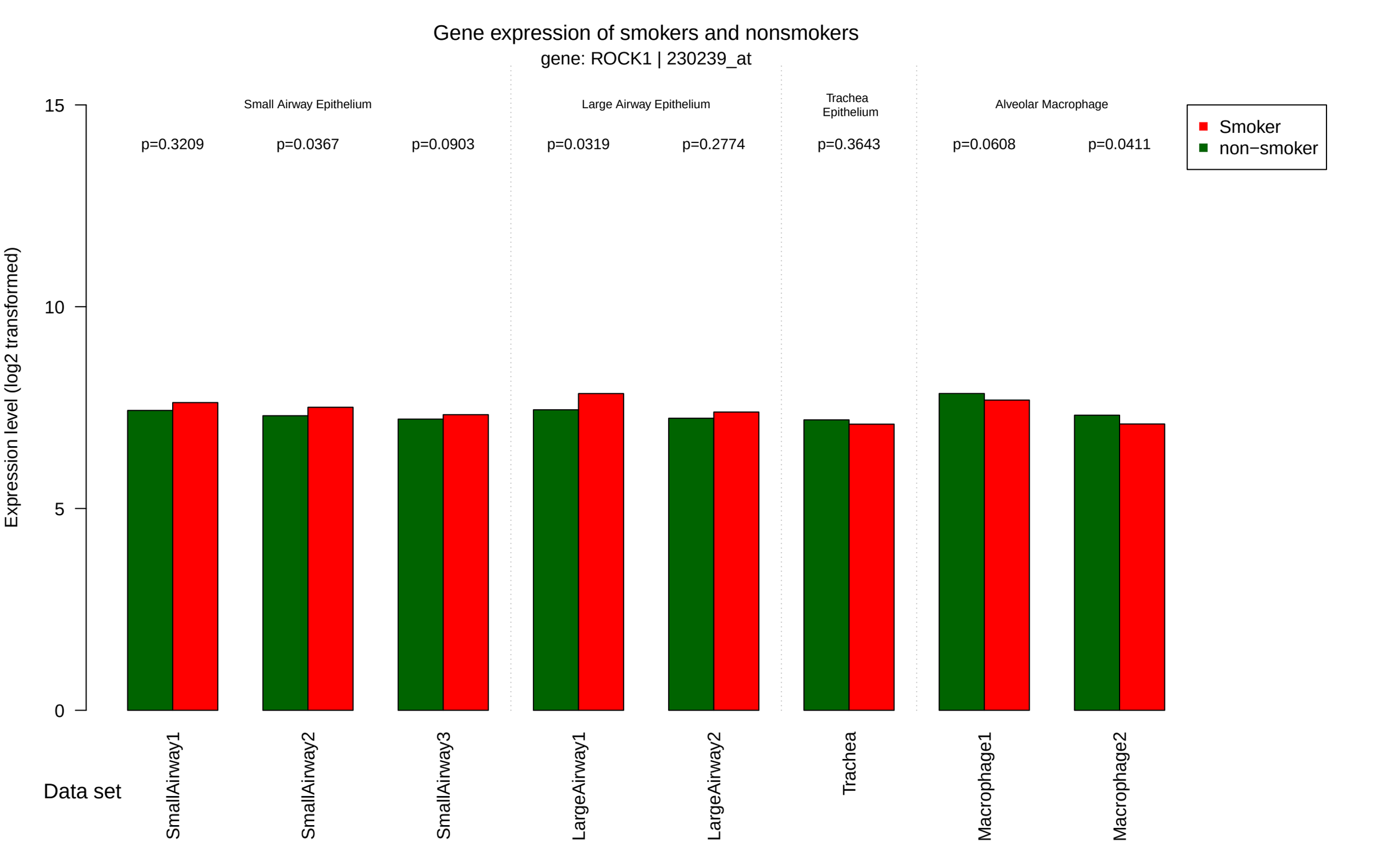
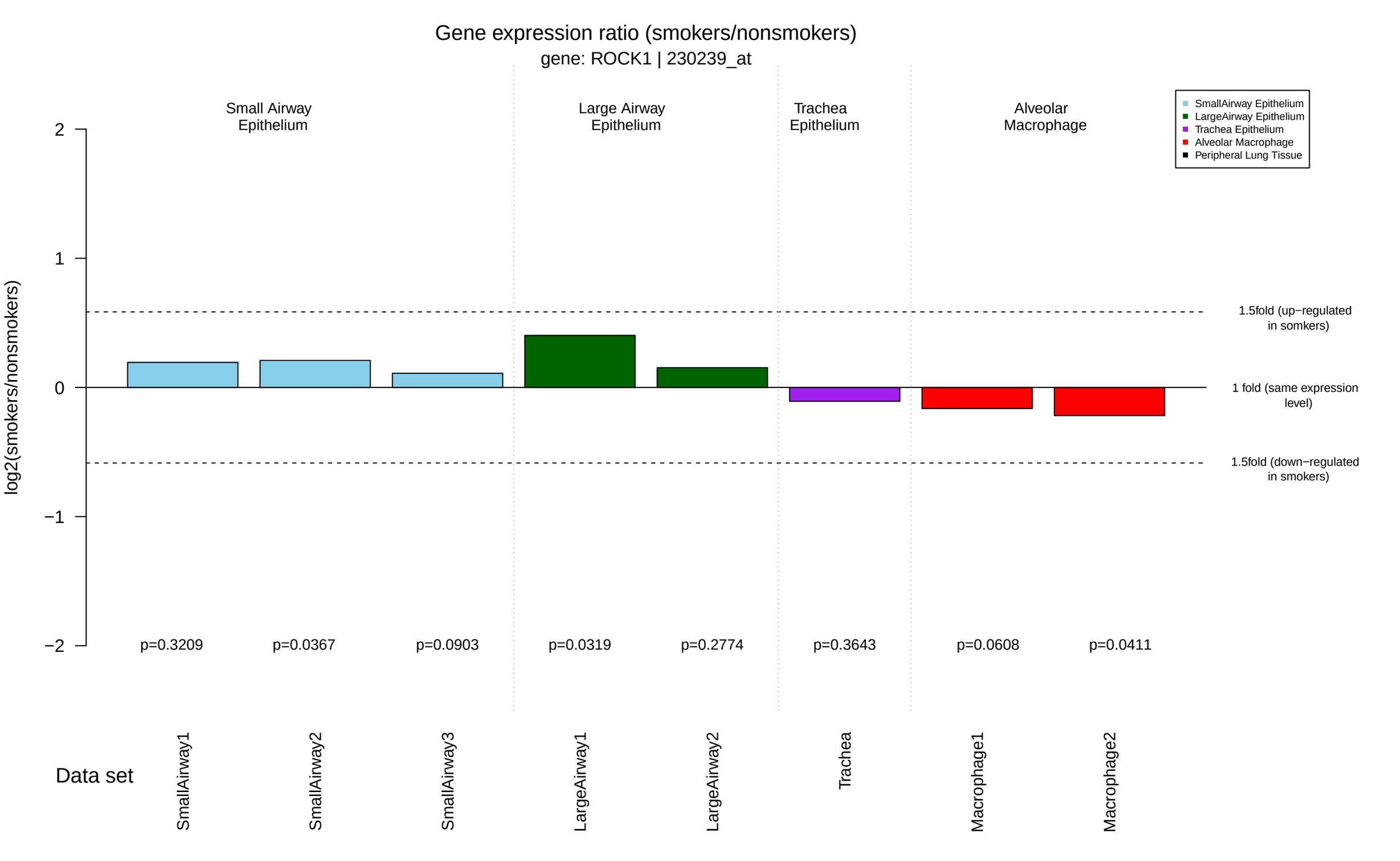
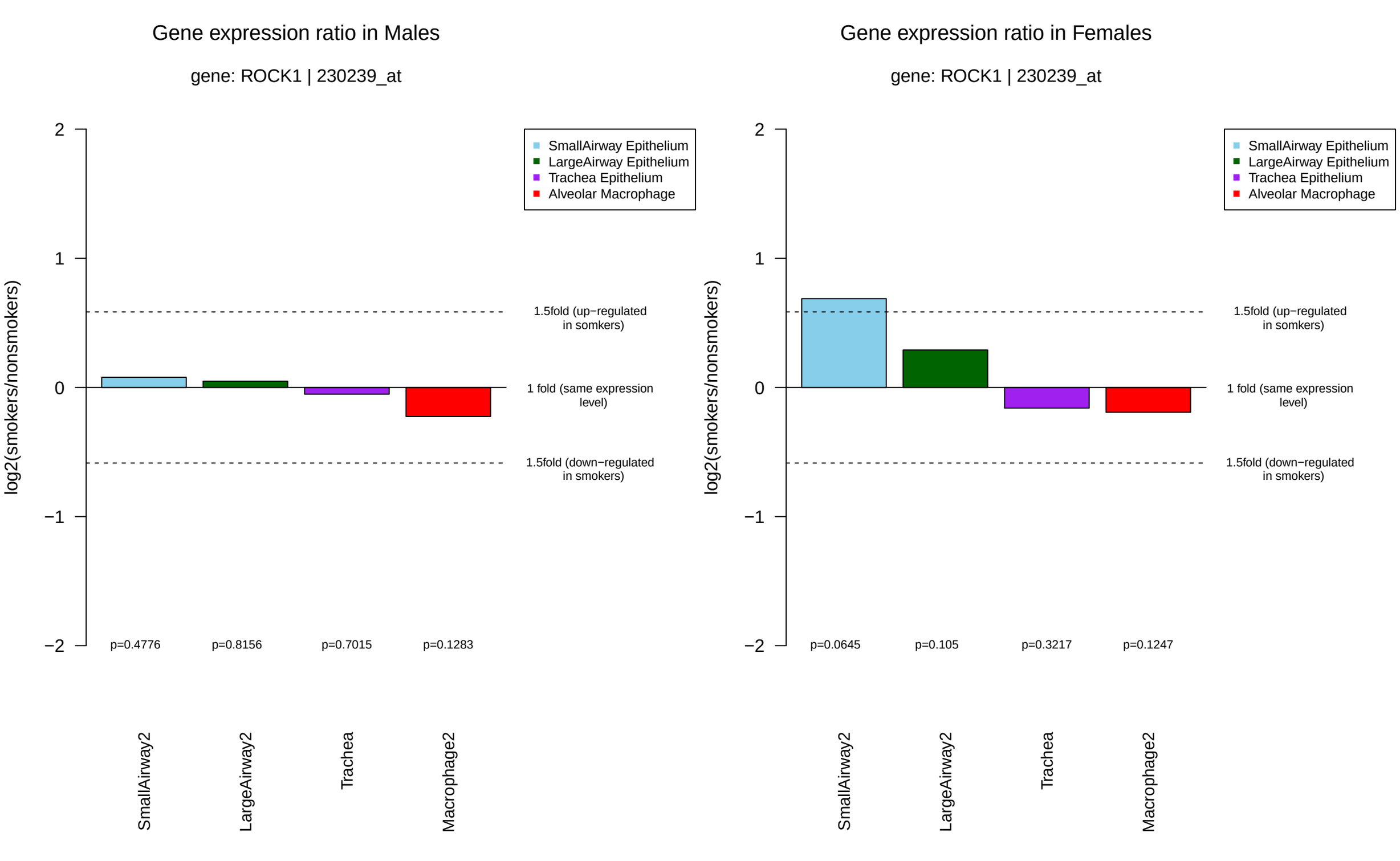
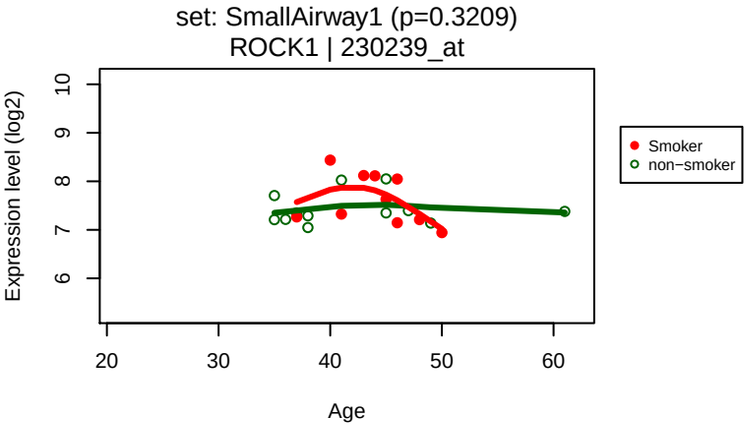
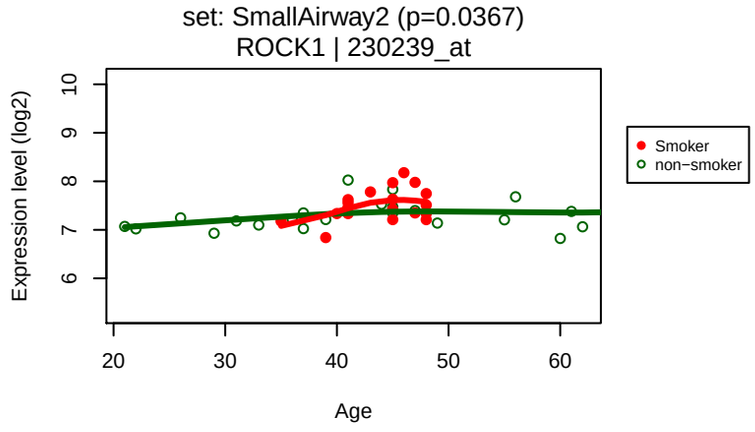
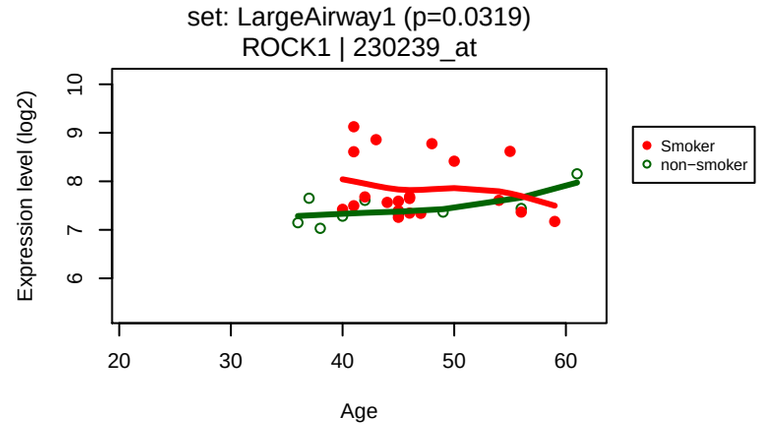

Gene expression of smokers and nonsmokers
gene: ROCK1 | 230239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3209
p=0.0367
p=0.0903
p=0.0319
p=0.2774
p=0.3643
p=0.0608
p=0.0411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

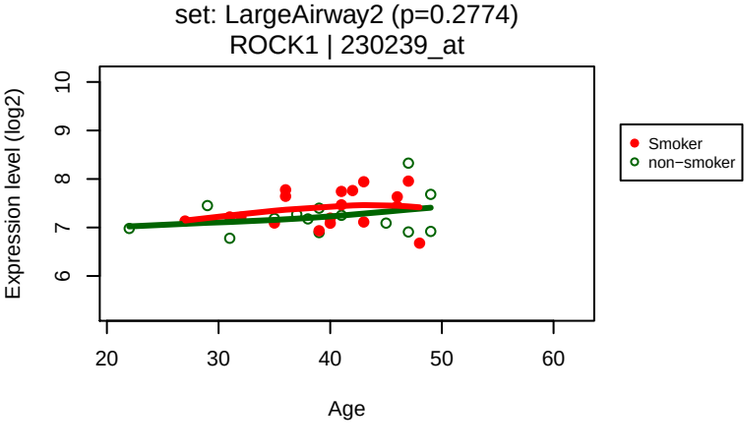

Gene expression ratio (smokers/nonsmokers)
gene: ROCK1 | 230239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3209
p=0.0367
p=0.0903
p=0.0319
p=0.2774
p=0.3643
p=0.0608
p=0.0411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

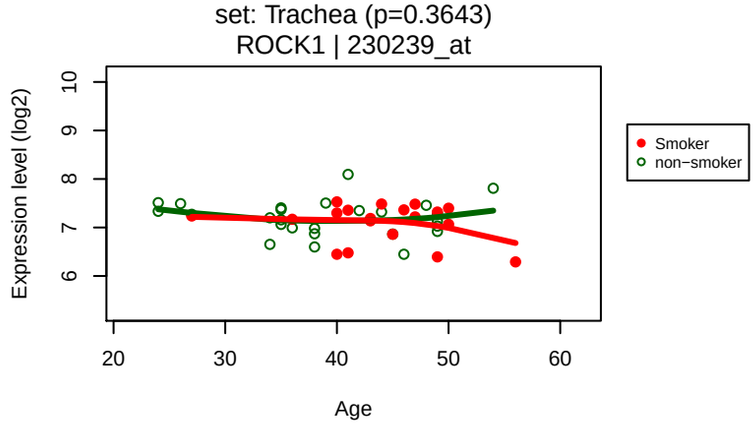

Gene expression ratio in Males
gene: ROCK1 | 230239_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4776
p=0.8156
p=0.7015
p=0.1283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROCK1 | 230239_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.105
p=0.3217
p=0.1247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

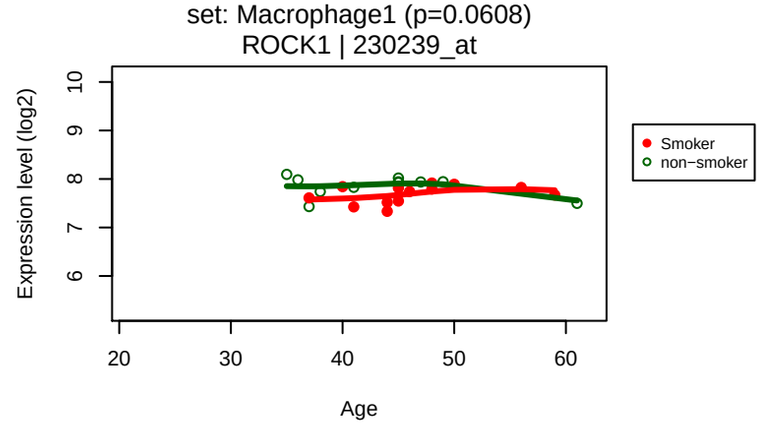

set: SmallAirway1 (p=0.3209)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker

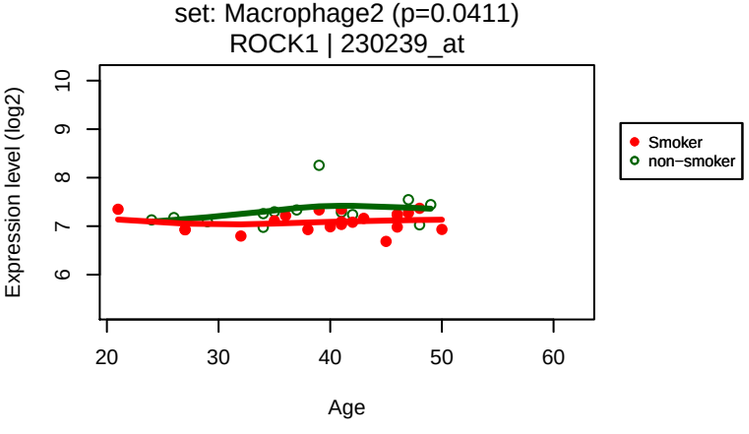

set: SmallAirway2 (p=0.0367)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker

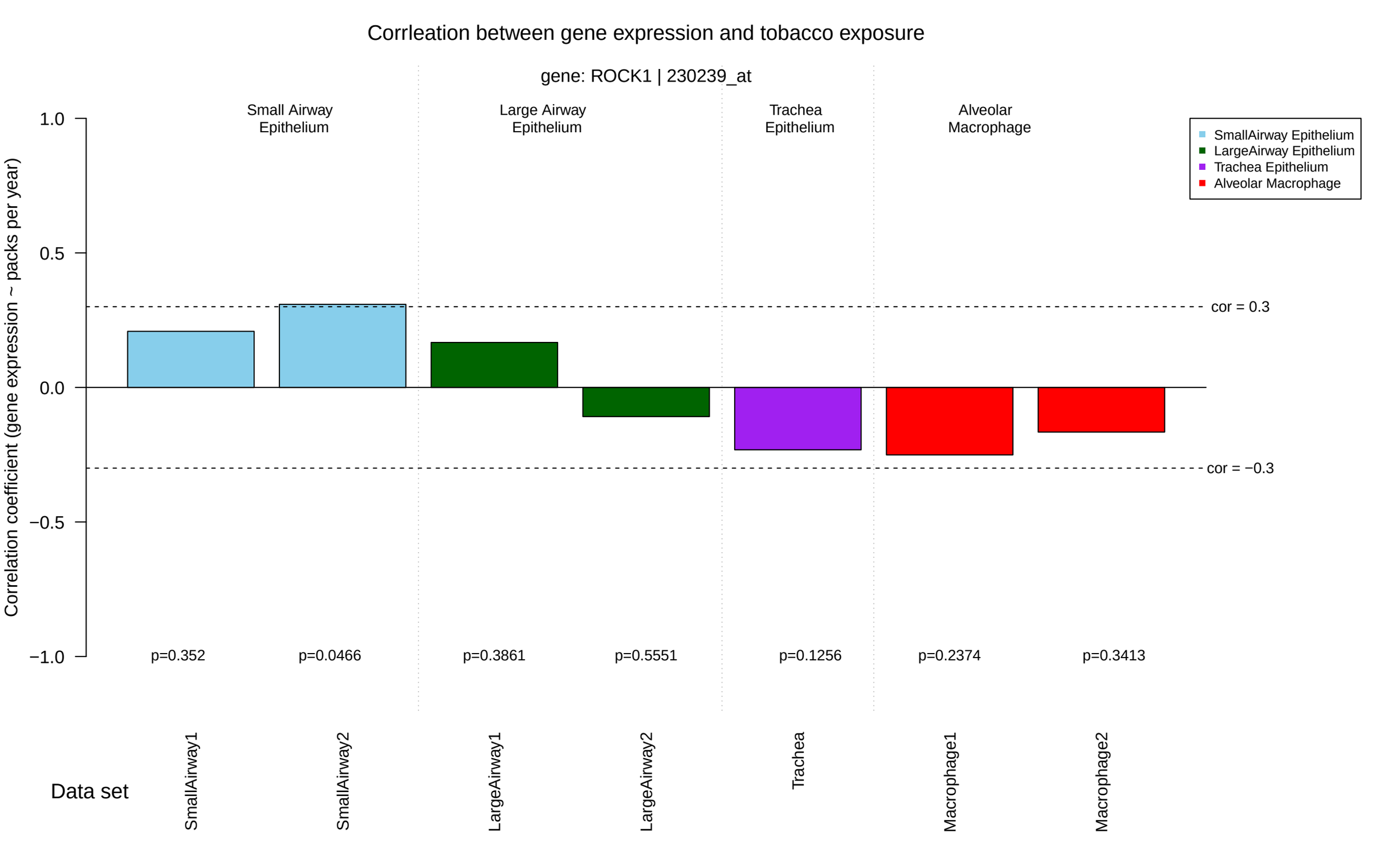

set: LargeAirway1 (p=0.0319)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2774)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3643)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0608)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0411)
ROCK1 | 230239_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROCK1 | 230239_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.352
p=0.0466
p=0.3861
p=0.5551
p=0.1256
p=0.2374
p=0.3413
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2